Regularization in Machine Learning

Multiple Linear Regression and Polynomial Regression

1 Connection with Previous Topics

We have already studied:

- Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Loss and Cost Function
- Gradient Descent
- Underfitting and Overfitting

In Polynomial Regression, increasing the degree of the polynomial makes the model more flexible.

For example,

$$\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3 + \cdots + w_n x^n$$

A very high-degree polynomial can fit the training data extremely well, but may perform poorly on new data.

This is called **overfitting**.

Key Concept

Overfitting occurs when a model learns the training data too closely and becomes too complex, resulting in poor performance on unseen data.

Therefore, we need a way to control the complexity of the model.

This leads to the idea of **Regularization**.

2 What is Regularization?

Regularization is a technique used to reduce overfitting by discouraging the model from using excessively large coefficients.

The basic idea is:

$$\boxed{\text{New Cost} = \text{Original Cost} + \text{Penalty}}$$

The penalty is related to the model coefficients.

Therefore, the model does not only try to minimize prediction error. It also tries to keep the coefficients under control.

Key Concept

Regularization does not remove features directly.
Instead, it adds a penalty to the cost function so that the model is discouraged from assigning unnecessarily large coefficients.

3 Why Penalize Large Coefficients?

Consider the following model:

$$\hat{y} = 5 + 8x_1 + 3x_2$$

Here,

$$w_1 = 8, \qquad w_2 = 3$$

The coefficient 8 is relatively large compared with 3.

Regularization adds a penalty for large coefficients.

Therefore, during training, the model has an additional reason to avoid unnecessarily large values.

The goal is not simply to make every coefficient zero.

The goal is to obtain a model that is sufficiently accurate while keeping the coefficients controlled.

4 The Role of λ

The symbol λ (lambda) controls the strength of regularization.

$$\boxed{\lambda = \text{regularization strength}}$$

The general idea is:

$$\lambda \uparrow \quad \Rightarrow \quad \text{stronger penalty}$$

and

$$\lambda \downarrow \quad \Rightarrow \quad \text{weaker penalty}$$

Important

A very small λ provides little regularization.
A very large λ may restrict the model too much and can lead to underfitting.
Therefore, λ must be chosen appropriately.

5 Simple Numerical Example

Suppose the model has two coefficients:

$$w_1 = 4, \qquad w_2 = 10$$

and the original cost is:

$$J = 5$$

Let

$$\lambda = 0.1$$

For Ridge regularization, the penalty is:

$$\lambda(w_1^2 + w_2^2)$$

Therefore,

$$\text{Penalty} = 0.1(4^2 + 10^2)$$

$$= 0.1(16 + 100)$$

$$= 11.6$$

Hence,

$$J_{\text{new}} = 5 + 11.6$$

$$\boxed{J_{\text{new}} = 16.6}$$

This shows that large coefficients contribute more to the penalty.

6 Ridge Regression

Ridge Regression uses **L2 regularization**.

The regularized cost function can be written as:

$$\boxed{J_{\text{Ridge}} = J + \lambda \sum_{j=1}^{n} w_j^2}$$

The additional term

$$\lambda \sum_{j=1}^{n} w_j^2$$

is the regularization penalty.

Key Concept

Ridge Regression

- Uses L2 regularization.
- Squares the coefficients in the penalty.
- Encourages coefficients to become smaller.
- Usually does not make coefficients exactly zero.

For example, suppose:

$$w_1 = 8, \qquad w_2 = 3$$

The Ridge penalty is:

$$\lambda(8^2 + 3^2)$$

$$= \lambda(64 + 9)$$

$$\boxed{73\lambda}$$

Thus, larger coefficients contribute more to the penalty.

7 Lasso Regression

Lasso Regression uses **L1 regularization**.

Its cost function is:

$$\boxed{J_{\text{Lasso}} = J + \lambda \sum_{j=1}^{n} |w_j|}$$

The penalty uses the absolute value of the coefficients.

Key Concept

Lasso Regression

- Uses L1 regularization.
- Uses absolute values of coefficients.
- Shrinks coefficients.
- Can make some coefficients exactly zero.
- Therefore, it can perform feature selection.

For example, suppose the model initially has:

$$w_1 = 5, \qquad w_2 = 3, \qquad w_3 = 0.2$$

After applying Lasso, an illustrative result could be:

$$w_1 = 4.2, \qquad w_2 = 2.5, \qquad w_3 = 0$$

Since

$$w_3 = 0$$

the corresponding feature no longer contributes to the prediction.

Important

Do not say that Lasso always removes features.
Lasso **can make some coefficients exactly zero**, which allows it to perform feature selection.

8 Ridge vs Lasso

	Ridge	Lasso
Regularization	L2	L1
Penalty	$\lambda \sum w_j^2$	$\lambda \sum \lvert w_j \rvert$
Main effect	Shrinks coefficients	Shrinks coefficients
Can make coefficient exactly zero?	Usually no	Yes
Feature selection	No	Yes

Key Concept

The simplest way to remember:

$$\boxed{\text{Ridge} \rightarrow \text{Shrink}}$$

$$\boxed{\text{Lasso} \rightarrow \text{Shrink} + \text{Possible Zero}}$$

9 Connection with Gradient Descent

We have already studied Gradient Descent as an optimization technique.
Recall:

$$\boxed{\text{Cost} \rightarrow \text{Gradient} \rightarrow \text{Parameter Update}}$$

Regularization changes the cost function.
Therefore, the gradient also changes.
The overall flow becomes:

$$\boxed{\text{Regularized Cost} \rightarrow \text{New Gradient} \rightarrow \text{New Parameter Updates}}$$

Thus, Gradient Descent can still be used to minimize the regularized cost function.

Important

Regularization does not replace Gradient Descent.
Gradient Descent is the optimization method, while regularization modifies the objective being optimized.

10 Overall Concept

The complete idea can be summarized as:

$$\boxed{\text{Polynomial Regression} \rightarrow \text{Model Complexity} \rightarrow \text{Overfitting} \rightarrow \text{Regularization}}$$

Regularization modifies the cost:

$$\boxed{\text{Original Cost} + \text{Penalty}}$$

Then:

$$\boxed{\text{Regularized Cost} \rightarrow \text{Gradient} \rightarrow \text{Parameter Updates}}$$

The two important approaches are:

$$\boxed{\text{Ridge} = \text{L2}}$$

$$\boxed{\text{Lasso} = \text{L1}}$$

11 Quick Revision

1. Why do we need regularization?
   To reduce overfitting and control model complexity.
2. What does regularization do?
   It adds a penalty to the cost function.
3. What does λ represent?
   The strength of the regularization penalty.
4. What is Ridge Regression?
   Ridge uses L2 regularization and shrinks coefficients.
5. What is Lasso Regression?
   Lasso uses L1 regularization and can make some coefficients exactly zero.
6. Does regularization replace Gradient Descent?
   No. It changes the cost function that Gradient Descent minimizes.

12 Key Takeaways

- Overfitting occurs when the model becomes too complex.
- Regularization helps control model complexity.
- Regularization adds a penalty to the cost function.
- λ controls the strength of the penalty.

- Ridge uses L2 regularization.
- Ridge generally shrinks coefficients toward zero.
- Lasso uses L1 regularization.
- Lasso can make some coefficients exactly zero.
- Gradient Descent can be used to optimize the regularized cost function.

Regularization = Better control over model complexity